

TEC-401

Roll No.

B. TECH. (ECE) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June/July, 2022

COMMUNICATION SYSTEMS—I

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the operation of the ring modulator circuit which generates the DSB-SC wave. (CO1)

(b) An amplitude modulated signal is given by :

$$s(t) = 10 \cos(2\pi 10^6 t) + 5 \cos(2\pi 10^6 t) \times \cos(2\pi 10^3 t)$$

$$+ 2 \cos(2\pi 10^6 t) \times \cos(4\pi 10^3 t) \text{ volt.}$$

Find the various frequency components present and the corresponding modulation indices. Draw the line spectrum and final bandwidth. (CO1)

P. T. O.

- (c) (i) Calculate the percentage power saving when the carrier and one of the sidebands are suppressed in an AM wave modulated to a depth of (1) 100 percent and (2) 50 percent.

- (ii) An AM modulator has output :

$$x(t) = A \cos 400 \pi t + B \cos 380 \pi t + B \cos 420 \pi t.$$

The carrier power is 100 W and the efficiency is 40%. Determine the value of A and B. (CO1)

2. (a) Explain with the help of block diagram, the Armstrong method of generating FM signal. (CO2)

- (b) Explain with necessary mathematical relations, the demodulation of an FM wave using PLL. (CO2)

- (c) An angle modulated signal with carrier frequency $\omega_c = 2\pi \times 10^5$ is described by the equation : (CO2)

$$x(t) = 10 \cos (\omega_c t + 5 \sin 3000 \pi t + 10 \sin 2000 \pi t).$$

- (i) Find the power of the modulated signal.

- (ii) Find the frequency deviation.

- (iii) Find the deviation ratio.

- (iv) Find the phase deviation.

- (v) Estimate the bandwidth of $x(t)$.

3. (a) Show that the r.m.s. value of the thermal noise voltage that appears

across the capacitor in a parallel R-C circuit is given by $\sqrt{\frac{kT}{C}}$. (CO3)

- (b) Derive an expression for the Friiss' noise figure formula. (CO3)
- (c) Derive the expression for noise equivalent bandwidth and establish the relation between noise equivalent bandwidth and 3 dB bandwidth for a RC low pass filter. (CO3)
4. (a) What is the necessity of pre-emphasis and de-emphasis used in FM system ? Explain pre-emphasis and de-emphasis with a neat circuit diagram. (CO4)
- (b) Show that in an FM system the figure of merit, assuming sinusoidal modulation, is given by $\gamma_{FM} = \frac{3}{2} m_f^2$, where m_f is the modulation index for FM. (CO4)
- (c) Assuming sinusoidal modulation, show that in an AM system with envelope detection the figure of merit is given by $\gamma_{AM} = \frac{\mu^2}{2 + \mu^2}$, where μ is the modulation index for AM. (CO4)
5. (a) With the help of a neat circuit diagram, explain the generation and demodulation of PWM. (CO5, CO6)
- (b) State and prove sampling theorem. (CO5, CO6)
- (c) Explain the need for modulation in a communication system. (CO5, CO6)

Roll No.

TEC-402

B. TECH. (ECE) (FOURTH SEMESTER) END SEMESTER EXAMINATION, June/July, 2022

ANALOG INTEGRATED CIRCUITS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) The following specifications are given for the dual-input, balanced-output differential amplifier where $R_c = 2.2 \text{ k}\Omega$, $R_E = 4.7 \text{ k}\Omega$, $R_{in1} = R_{in2} = 50 \text{ }\Omega$, $+V_{cc} = 10 \text{ V}$, $-V_{EE} = -10 \text{ V}$ and the transistor $\beta_{dc} = \beta_{ac} = 100$ and $V_{BE} = 0.715 \text{ V}$ typical : (CO1)

(i) Determine the ICQ and VCEQ.

(ii) Determine the voltage gain.

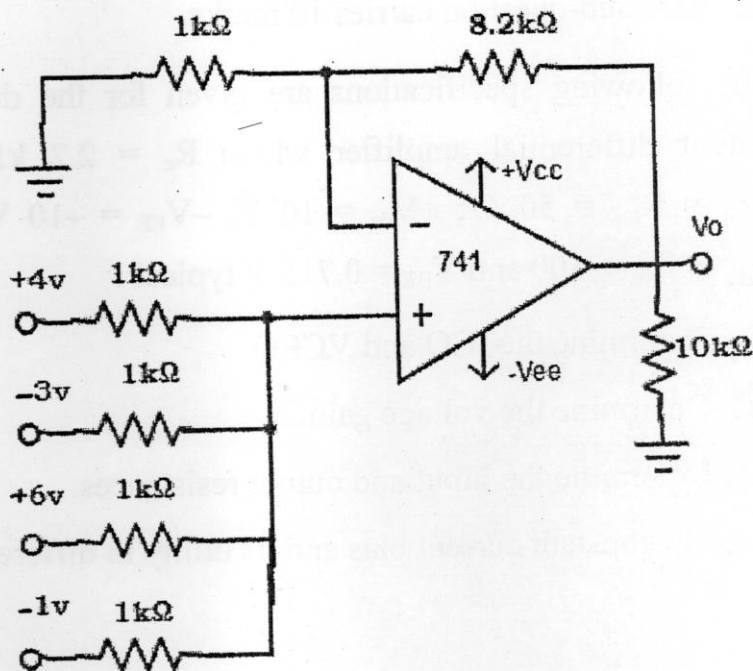
(iii) Determine the input and output resistances.

- (b) Explain constant current bias and its utility in differential amplifier.

(CO1)

P. T. O.

- (c) Explain Wildar Current source. Design a Wilder current source for generating a constant current $I_o = 10 \mu\text{A}$. Assume $V_{cc} = 10 \text{ V}$, $V_{BE} = 0.7 \text{ V}$, $\beta = 125$. Use $V_T = 25 \text{ mV}$ and $I_{ref} = 1 \text{ mA}$. (CO1)
2. (a) Explain the DC Characteristics of OPAMP. (CO2)
- (b) The 741 C OP-Amp having the following parameters is connected as a inverting amplifier with $R_i = 1 \text{ k}\Omega$ and $R_f = 10 \text{ k}\Omega$, $A = 200000$, $R_i = 2 \text{ M}\Omega$, $R_o = 75 \Omega$, supply voltages of magnitude 15 V and output voltage swing is $\pm 13 \text{ V}$. Calculate the values of A_F , R_{iF} , R_{oF} and bandwidth. (CO2)
- (c) Explain Voltage to current Converter with example. (CO2)
3. (a) Explain, how instrumentation amplifier is different from ordinary op-amp and explain instrumentation with linear gain control. (CO3)
- (b) Explain Averaging Amplifier. Find the value of V_1 in the circuit shown below. : (CO3)



(3)

- (c) Design an op-amp differentiator that will differentiate an input signal with $f_{\max} = 100$ Hz. Draw the output waveform for a sine wave of 1 V peak at 100 Hz applied to the differentiator and apply same for square wave input. (CO3)
4. (a) Explain half wave and full wave rectifier with superdiodes. (CO4)
- (b) Explain the compensating LOG amplifier. (CO4)
- (c) Explain positive feedback comparator with positive reference. In Schmitt trigger $R_2 = 100 \Omega$, $R_1 = 50 \text{ k}\Omega$, $V_{\text{ref}} = 0 \text{ V}$, $V_i = 1 \text{ Vpp}$ sine wave and saturation voltage = $\pm 10 \text{ V}$. Determine threshold voltages V_{UT} and V_{LT} . (CO4)
5. (a) Explain state variable filter and how it shows the various filter characteristics. (CO5 and CO6)
- (b) Design a 2nd order Butterworth filter using sallen key unity gain low pass filter of critical frequency 3 kHz and passband gain of 4. (CO5 and CO6)
- (c) What is the function of a voltage regulator ? State advantages and disadvantages of a series voltage regulator. Explain series OP-AMP regulator. (CO5 and CO6)

Roll No.

TEC-403

B. TECH. (EC) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June/July, 2022
MICROPROCESSOR AND ITS APPLICATIONS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Draw and explain block diagram of 8259. (CO3, CO4)
(b) Explain the difference between : (CO3, CO4)
 - (i) AAA and AAS instruction
 - (ii) DAA and DAS
 - (iii) NOT and NEG(c) Write 8086 assembly language program to transfer 15 bytes of data from location 2050 to 3050 using string instructions. (CO3, CO4)
2. (a) What is Interrupt ? Classify interrupts in 8085 ? (CO1, CO2)
(b) Write a program to generate a delay of 10 ms in 8085. (CO1, CO2)
(c) Write 8085 assembly language program to convert a Binary number to its BCD equivalent. (CO1, CO2)

P. T. O.

(2)

3. (a) Explain concept of memory interfacing with the example to interface four 16K RAM consecutively with microprocessor 8085. (CO3, CO4)
- (b) What are the different addressing modes in 8086 ? Explain with example. (CO3, CO4)
- (c) Explain the functions of following 8086 pins : (CO3, CO4)
- (i) HLDA
 - (ii) READY
 - (iii) DEN
 - (iv) QS0, QS1
4. (a) Draw Block diagram of Intel 8086 Microprocessor. Explain the concept of Pipelining. (CO4)
- (b) Write a 8086 assembly language program to convert a Binary number to its GRAY code equivalent. (CO4)
- (c) Explain the meaning of the following 8086 instructions : (CO4)
- (i) ADD [4500], BX
 - (ii) MOVSB
 - (iii) MOV BX, 1234
 - (iv) XOR AX, [BX]
5. (a) Draw and explain block diagram of 8254. (CO5, CO6)
- (b) Program 8255 for the following configurations : (CO5, CO6)
- (i) Port A used as Input port and Port C as output port.
 - (ii) To get data from Port B and send its Gray code to port C.
- (c) Explain the working of 8255 in BSR mode. (CO5, CO6)

Roll No.

TEC-404

B. TECH. (ECE) (FOURTH SEMESTER)

END SEMESTER EXAMINATION, June/July, 2022

ELECTROMAGNETIC FIELD THEORY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Given vectors $T = 2a_x - 6a_y + 3a_z$ and $S = a_x + 2a_y + a_z$. Determine

(i) the scalar projection of T on S, (ii) the vector projection of S on T and (iii) the smaller angle between T and S. (CO1)

(b) Express the following vector in Cartesian coordinates : (CO1)

$$\vec{A} = \rho(z^2 + 1)\hat{a}_\rho - \rho z \cos \phi \hat{a}_\phi$$

(c) Determine the electric field due to the following potential : (CO1)

(i) $V = \rho^2(z + 1) \sin \phi$

(ii) $V = e^{-r} \sin \theta \cos 2\phi$

P. T. O.

2. (a) Explain the shortcoming in Ampere's law and modification made by Maxwell in Ampere's law. (CO2)

(b) Explain the Gauss' law. Let $\vec{D} = 2\rho z^2 \hat{a}_\rho + \rho \cos^2 \phi \hat{a}_z$. Determine the charge enclosed by the region $0 \leq \rho \leq 5, -1 \leq z \leq 1, 0 < \phi < 2\pi$. (CO2)

- (c) A charge distribution with spherical symmetry has density :

$$\rho_v = \begin{cases} \frac{\rho_0 r}{R}, & 0 \leq r \leq R \\ 0, & r > R \end{cases}$$

Determine E everywhere.

(CO2)

3. (a) Write the Maxwell's equation in most generalized form (both differential and integral), charge free region (both differential and integral) and steady field (both differential and integral). (CO3, CO6)

(b) Show that the tangential component of electric field strength vector and normal component of electric flux is continuous across the boundary interface. (CO3, CO6)

(c) Region 1 ($z < 0$) contains a dielectric for which $\epsilon_r = 2.5$, while region 2 ($z > 0$) is characterized by $\epsilon_r = 4$. Let $\vec{E}_1 = -30\hat{a}_x + 50\hat{a}_y + 70\hat{a}_z$ V/m. Find (i) D_2 , (ii) the angle between E_1 and the normal to the surface. (CO3, CO6)

4. (a) Explain quarter wave transformer in transmission line and its applications. A lossless transmission line operating at 4.5 GHz has $L = 2.4 \mu\text{H/m}$ and $Z_0 = 85 \Omega$. Calculate the phase constant β and phase velocity u . (CO4, CO6)

(3)

- (b) What are the conditions of lossless and distortionless transmission line ?
A distortionless line operating at 120 MHz has $R = 20 \Omega/\text{m}$, $L = 0.3 \mu\text{H}/\text{m}$ and $C = 63 \text{ pF}/\text{m}$. Determine γ , u and Z_0 . (CO4, CO6)
- (c) Define the term polarization for a uniform plane wave propagating in free space. Explain different types of polarizations in brief. (CO4, CO6)
5. (a) What do you understand by TE, TM and TEM mode of parallel plate waveguide ? Draw the field lines for TEM, TE_1 and TM_1 for parallel plate waveguide. (CO5)
- (b) If $\vec{H} = 10 \sin(\omega t - 4z) \hat{a}_x$, mA/min. a material for which $\sigma = 0$, $\mu = \mu_0$ and $\epsilon = 4\epsilon_0$, calculate ω , λ and $\mathbf{J}_d$. (CO5)
- (c) Consider an infinitely long transmission line consisting of two concentric cylinders having their axes along the z -axis. The inner conductor has radius a and carries current I while the outer conductor has inner radius b and thickness t and carries return current $-I$. Determine $\vec{H}$ everywhere and plot the magnitude of $\vec{H}$ with radius of cylinder. (CO5)

Roll No.

TEC-451

B. TECH. (ECE-ESR) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June/July, 2022

INTRODUCTION TO ROBOTICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the working of robot manipulator and end effector. (CO1)
- (b) Describe and explain major components of robot in detail. (CO1)
- (c) Explain the *three* laws of robotics. Differentiate between sensors and actuators with examples. (CO1)

P. T. O.

2. (a) Derive an expression for unit step response of second order system. (CO2)
- (b) Explain Linear Variable Differential Transformer (LVDT). (CO2)
- (c) What is the effect of the "integral" component in PID control ? (CO2)
3. (a) Write a short note on grippers and explain its various types in detail. (CO3)
- (b) Explain Hall effect sensors in detail with a proper diagram. (CO3)
- (c) Assume that a positioning measuring potentiometer (with total resistance $R = 15.5 \text{ K}\Omega$, total length $L = 22 \text{ cm}$ and input voltage $V_i = 10 \text{ V}$) has the wiper at $x = 12 \text{ cm}$. The voltage V_o of potentiometer is measured with an analog voltmeter with $R_o = 35 \text{ K}\Omega$ connected as load of potentiometer. (CO3)
 - (i) Calculate V_o measured by analog voltmeter.
 - (ii) Calculate the error in analog voltmeter of V_o with regard to V_o measured by digital voltmeter with $R_o = 12 \text{ M}\Omega$.
4. (a) Determine the position and orientation of 2-link manipulator. (CO4, CO6)
- (b) Explain about forward and inverse kinematics of manipulator in detail. (CO4, CO6)
- (c) Prove the given expression : (CO4, CO6)

$${}^F R_{M1} = {}^F R_M \times {}^M R_{M1}$$

(3)

5. (a) A point $a_{uvw} = (4, 3, 2)$ is attached to a rotating frame, which was obtained by rotating 60 degrees about OZ-axis of the reference frame (x, y, z -frame). Find the coordinates of the point relative to the reference frame. (CO4, CO5)
- (b) Write a short note on Homogeneous Transformation. (CO4, CO5)
- (c) Explain about sensing and digitizing function in machine vision.

(CO4, CO5)

Roll No.

TEC-491

B. TECH. (ECE) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June/July, 2022
SENSORS AND SIGNAL CONDITIONING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain Interfering and modifying inputs and configuration techniques of measurement system. (CO1)
- (b) What do you understand by Accuracy, Precision and Sensitivity of measurement system ? (CO1)
- (c) With the help of suitable example explain zero order dynamic characteristics of the system. (CO1)
2. (a) Explain in brief about the working principle and applications of thermistor. (CO2)
- (b) Explain in brief about force sensor and torque sensor. (CO2)

P. T. O.

(2)

- (c) What are the dielectric materials and also explain their uses in sensors ?
(CO2)
3. (a) What do you understand by Loading effects on sensor's output ? (CO3)
(b) With the suitable diagram explain RC band reject filter and find their parameters. (CO3)
(c) Design a basic wide band RC band stop filter with lower cut off frequency of 200 Hz and a higher cut off frequency of 800 Hz. Find geometric center frequency, bandwidth and Q point of the Filter. (CO3)
4. (a) Find out the expression for closed loop gain of inverting amplifier.
(CO4)
(b) What do you understand by current to voltage converter ? (CO4)
(c) What are the various applications of Op-Amp in signal conditioning ?
(CO4)
5. (a) Explain first order filters and give suitable examples of first order filters. (CO5/CO6)
(b) Explain in brief about switch capacitor filter. (CO5/CO6)
(c) What do you understand by filter approximation ? (CO5/CO6)

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Paper Code: : TOE 410

END SEMESTER Examination

B.Tech (IT/EC) IV

Data Structure using 'C' language.

Time : Three Hours

Maximum Marks :100

INSTRUCTIONS TO STUDENTS

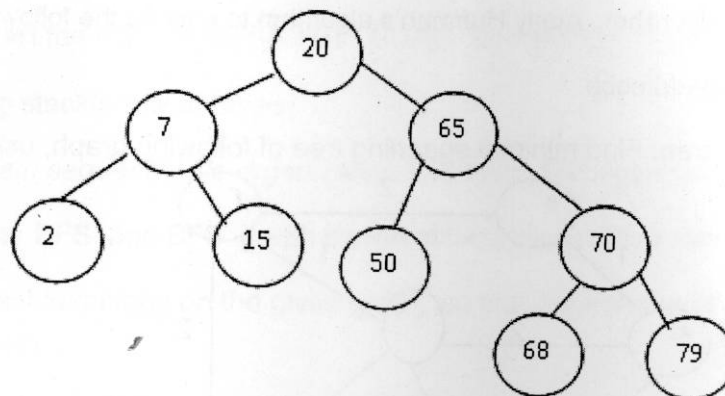
Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.

Q1.)

(2X10=20 Marks) (CO2, CO3, CO5)

- a. Explain binary tree and binary search tree. Write a 'C' function to create a binary search tree (do not use global variables).
- b. Draw the expression tree for the expression $(7+A/B)^3 * (2*C+D)^3$
- c. Consider the tree given below :



- i) Find the depth of the tree.
- ii) Draw the tree after insertion of node 66 showing each step.
- iii) Draw the tree after deletion of node 70 showing each step.

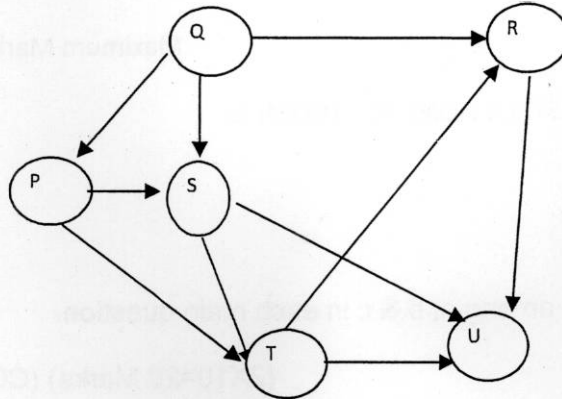
Q2.)

(2X10=20 Marks) ((CO3, CO4, CO5))

a. Explain B and B⁺ tree. Draw a B - tree of order 4 with following keys :

g, p, a, x, f, b, k, h, m, e, s, r, c, l, n.

b. Give linked representation and memory representation of following graph



c. Write a c function to count total number of nodes in a circular singly linked list.

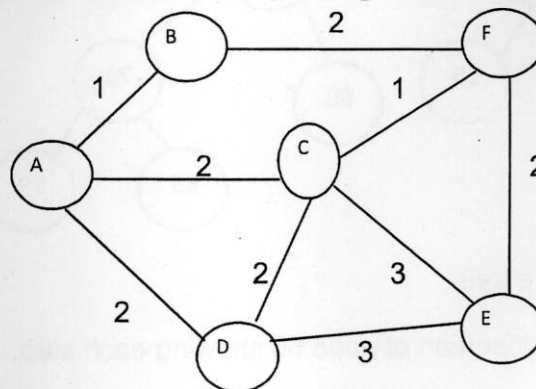
Q3.)

(2X10=20 Marks) (CO2, CO4, CO3)

a. Explain Huffman's algorithm. Apply Huffman's algorithm to encode the following signal.

aabadedddcffeafecabeddfecbb

b. Explain spanning tree. Find minimal spanning tree of following graph, using Kurskal's algorithm.



c. Explain hash collision with an example. Consider a hash table of size (m) 10. Using linear probing technique insert following keys 42,27,36,34,83,81,92,33 and 111 into the table.

Q4.)

(2X10=20 Marks) ((CO1, CO3, CO4)

- What do mean by an AVL tree? Write advantages of an AVL tree. Draw an AVL tree with following keys: 7, 5, 4, 8, 19, 3, 2, 20, 21, 1.
- Explain sorting and its types. Write a C function to sort an array using Merge sort technique.
- (i) What is the output of following code :-

```
main()
{
    int q, *p, n;
    q = 220;
    p = &q;
    n = *p;
    printf("%d", n);
}
```

- Show how the following polynomial can be represented using a linked list.

$$7x^2y^2 - 4x^2y + 5xy^2 - 2$$

Q5.

(2X10=20 Marks) (CO3, CO4, CO5)

- Convert the expression $((A + B) * C - (D * E) ^ (F / G))$ to its equivalent postfix expression using stack(show all steps).
- Explain sequential file organization and index file organization with examples
- Define DFS and BFS graph traversal techniques. Give name and apply a graph traversal technique on the given graph, so that the number of nodes between A to H are minimum.

